Predicting Emergency Department Fast-Track Eligibility

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ABSTRACT

A common approach to improve emergency department (ED) patient flow is the usage of fast-track (FT) areas, a special section within the ED staffed by nurse practitioners, physician assistants, or other healthcare staff not in the main ED area. FTs allow staff to promptly address patients with a lower triage acuity, preventing the waiting room from getting congested, and preventing long stays for patients with minor conditions that may be bypassed by those with more critical conditions. With machine learning, incoming patient data could aid triage staff in making a quicker determination whether a patient should be fast-tracked. This paper presents a model capable of making a fast-track recommendation for a patient coming into the ED based on commonly provided patient data such as age, gender, race, initial vital readings, as well as relevant free-text data like chief complaint, home medications, and past medical history through the use of different BERT embeddings. Data sourced from a diverse set of ED visits provided by the MC-MED dataset were used for training and evaluation, with performance on the MC-MED test set showing an AUC-ROC score of around 0.82. In future work, this model and could be extended to make use of additional multi-modal data or even be further built upon for deployment to promote fast-track eligible patients to forgo the ED for urgent care or outpatient facilities.

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1 INTRODUCTION

In a hospital emergency department (ED), time is critical [35]. If a patient with symptoms such as serious chest or abdominal pain is not seen right away, their situation has the potential to quickly worsen and become life threatening. Since many EDs are open 24 hours a day, 7 days a week to any patient seeking care, ED occupancy becomes an important metric to determine the overall patient flow into and out of the department, and gives a solid picture of overall ED performance [23]. One proven strategy to mitigate overcrowding is the introduction of a fast-track (FT) system within the ED, a separate segment of the ED that targets patients with lower or medium acuity that have less urgent care needs, or that meet some other criteria set by the triage staff such as the presence of specific comorbidities. FT areas are staffed independently of the main ED or other specialized areas, and sometimes have nurse practitioners, junior doctors, or other mid-level staff rather than physicians, and the care impact of which role is still being studied [29]. However with documented positive impacts to length of stay (LOS), patient satisfaction, and other important ED metrics [6, 13, 25, 29], it is clear why fast-track areas have become more common in hospitals across the US [19].

Hospitals stand to benefit from systems that could aid triage staff in more quickly making FT or non-FT decisions. Triage algorithms such as the Emergency Severity Index (ESI) require staff to make triage decisions with a flow chart based on answers to specific questions, however since the ED is a dynamic environment where incoming patient load cannot be reliably estimated, any tools or systems capable of reducing the time it takes staff to keep patients moving in and out of the ED benefits the entire system [23].

Machine learning models are capable of training on vast amounts of information and noticing patterns or indicators of certain outcomes [30, 34]. In this paper, we present a deep neural network that can predict whether an incoming ED patient ought to be fast tracked or not using multi-modal information commonly gathered at patient arrival. Raw patient numerical data that are structured such as vitals, age, and race and are combined with calculated data such as Modified Early Warning (MEWS) score, total medications at visit, shock index, and then finally combined with natural language data like chief complaint, past medical history descriptions, and home medication lists in the form of BERT embeddings to provide a model that has a detailed snapshot of the patient entering the ED. The diversity of the input data is significant since raw numerical data such as vitals offers a limited view into the severity of the patient's condition for triage assessment [5]; triage staff generally have much more context available. Real de-identified patient data from the Stanford Health Care Emergency Department provided in the MC-MED dataset [14] were used to develop the model, with 5-level Emergency Severity Index (ESI) triage acuity scores serving as the ground truth as to whether the patient would have been fast-tracked. After feature selection and training, the model was able to achieve 0.82 ROC-AUC on the MC-MED test set with BioClinical BERT [3] embeddings giving the highest boost to accuracy of the BERT variants tested.

2 RELATED WORK

There have been multiple studies investigating the impact of machine learning models in hospital and ED settings, with some recent studies focusing on the impact of the usage of large language models such as ChatGPT for use cases such as triage and preliminary diagnosis [26, 33, 36]. One such paper focused on the impact of LLM-based triage decisions and found that ChatGPT was able to perform on a similar level to the original triage team [26]. They had a staffed triage team making decisions while in parallel an ED physician prompted the model with vitals, condition, medical history, and symptoms of the same patient. Then both triage labels were compared against an independent ED specialist determination with both the model and the staff achieving a similar AUC. The paper raised valid concerns about the privacy of patients when feeding data into a privately owned model and chose to explicitly leave out identifying information from prompts.

Some exploration has been done in understanding how multiple modalities affect healthcare [32]. The authors propose a framework that takes in patient data modalities such as tabular, time series, natural language, and xrays to determine how the addition of each modality affects the AUC. Our paper focuses only on raw tabular and natural language data. Time-series and xray data are also

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available in the MC-MED dataset we use for training, but were excluded since a fast-track decision would likely have been made by staff before the collection of these modalities, especially when considering the importance of time management in the ED [23].

There has been only limited research into using machine learning models specifically for fast-track decision-making. One paper from 2023 built a similar model by utilizing the ED data in the MIMIC IV dataset to evaluate how well a fast-track determination could be made across various types of models [16]. A key aspect to note is that they used acuity as a feature in training, and considered a fast-track case any LOS shorter than four hours. Our paper chose to exclude acuity as a feature and instead use that data as the target since acuity score is likely correlated with other fast-track variables such as vitals or chief complaint. In regards to their fast-track criteria, we wanted to make sure that high acuity patients would not be considered for fast-track, even with a short length of stay since the fast-track area is typically reserved for non-urgent patients [29]. They found that multi-layer perceptron (MLP) algorithms outperformed other all models tested. We were able to find many other studies relating to fast-track [6, 9, 18, 25, 29] but no others beside the one referenced that apply machine learning algorithms for patient fast-track prediction.

This paper combines both the language-oriented transformer approach through the use of different BERT embeddings with a more classic MLP-based approach with tabular features to give the model a diverse set of context about the patient, and is the first example that we could find of a multi-modal approach to fast-track prediction.

3 METHODS

3.1 Dataset

In this paper for training and evaluation we use the Multimodal Clinical Monitoring in the Emergency Department (MC-MED) dataset, a comprehensive ED dataset that contains real de-identified patient data from the Stanford Health Care Emergency Department from 2020 to 2022 [15]. The data are available on the PhysioNet platform only through credentialed access, requiring specimens-only research training and a data-use agreement before access is granted [10].

The data are composed of multiple csv files with information on the ED visit level such as demographics, vitals, home medications, past medical history, ordered labs, imaging, and both time-series and segmented waveform data for continuously monitored vitals. This paper focuses on patient data in the time between arrival and admission to the ED, so labs, imaging, and waveform data were excluded. The data are provided for 118,385 adult visits from 70,545 unique, 18-and-older patients with the original patient medical record numbers (MRN) and Contact Serial Numbers (CSN) remapped onto random integers. The data are additionally pre-split randomly into train, validation, and test sets that keep patients within the same set, and there is also an additional group of splits that ensures chronological order, but it was not needed in our research. The train set is composed of 94,656 visits, the validation 12,061 visits and the test set 11,665 visits. All dates in the dataset are shifted consistently for each patient but preserve the original seasonality. Due to the shifting, ED load during a particular period of time is not determinable, nor is staffing or resource allocation.

Table 1: MC-MED Dataset Demographics

Patient Characteristics (n=70,545)	
Age, years	
Mean ± SD	52.0 ± 20.4
Median [IQR]	51 [34.0-69.0]
≥ 65	21,723 (30.7%)
≥ 90	2,246 (3.1%)
Gender	
Female	37,941 (53.7%)
Male	32,569 (46.1%)
Race	
White	28,913 (41.1%)
Other/Unknown	23,329 (33.1%)
Asian	12,592 (17.9%)
Black	3,557 (5.0%)
Ethnicity	
Non-Hispanic	51,511 (73.2%)
Hispanic/Latino	18,012 (25.6%)
Visit Characteristics (n=118,385)	
Means of arrival	
Self	93,939 (79.2%)
EMS	24,313 (20.5%)
Triage Vitals	(median [IQR])
Temp (C)	36.7 [36.5-36.8]
Heart Rate (bpm)	84.0 [73.0-97.0]
Resp Rate (breaths per minute)	18.0 [16.0-18.0]
O2 Saturation	99.0 [98.0-100.0]
Systolic blood pressure (mmHg)	135.0 [121.0-149.0]
Diastolic blood pressure (mmHg)	79.0 [70.0-88.0]
Triage acuity (ESI)	
1-Resuscitation	1,066 (0.9%)
2-Emergent	27,880 (23.6%)
3-Urgent	77,259 (65.5%)
4-Less Urgent	10,973 (9.3%)
5-Non-Urgent	680 (0.5%)
Length of stay	
ED LOS, hours (median [IQR])	5.2 [3.5-7.2]
Hospital LOS, days (median [IQR])	3.0 [1.0-6.0]

3.2 Feature Selection

Features were broken up into three different groups: raw visit data in the form of tabular numerical or categorical data, calculated features based off of combined or transformed raw data, and free-text data processed into BERT embeddings. Through python scripts and the pandas library we generated a new csv for each group before running them through the final model architecture for training.

Raw data were captured directly from the MC-MED visit data and through custom python scripts and the use of the pandas library [22]. From the visit data we extracted number of visits, age, and triage vitals which included temperature, heart rate, respiratory rate, oxygen saturation, systolic blood pressure, and diastolic blood pressure. Additional categorical columns were one-hot encoded for

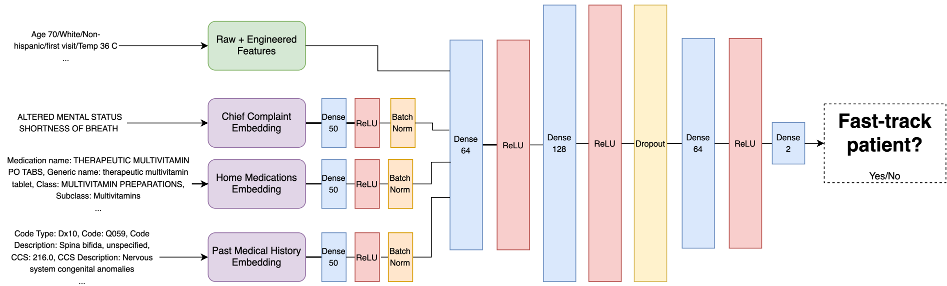

Figure 1: Fast-track prediction model architecture

race, ethnicity, means of arrival, gender, and payor class. It was important that we take the triage readings and not the acuity, orders, or other waveform vital data since triage staff would likely not have this data available before a fast-track determination because of items like waveform relying on bedside medical devices or orders relying on physician or nurse evaluation. Any missing non-categorical raw feature values were filled with median values rather than discarding the visit, so as not to lose other potentially rich patient information.

In this model the engineered or transformed columns included visit arrival time, visit arrival season, shock index, modified early warning score (MEWS), and total medication count. Arrival time only includes the time component and not a full date timestamp since original dates have been shifted. This value was made cyclical rather than linear so that 11:59 would still be close in value to 00:01. Shock index is a simple calculation of the heart rate divided by the systolic blood pressure that, when greater than one, can be tied to mortality other serious patient conditions [17]. MEWS takes systolic blood pressure, heart rate, respiratory rate, temp and consciousness, and assigns a point value from 1-3 based on the range of the vital. The score has been shown to indicate critical illness and other risks of health deterioration [8]. The consciousness rating was fixed at 0, meaning alert, for the MEWS score since in MC-MED this information is not available for a visit. It is also important to note that total medications is based on the start and end date of the patients medications but for some patients start and end dates are missing so medications were only included if they had a start before the arrival time, with end date causing exclusion only if present and before arrival.

In order not to lose potentially critical information about chief complaint, past medical history, and home medications, the patient natural language data were transformed into BERT embeddings for input into the model. Four BERT models were tried for embedding generation: BioClinicalBERT [2], PubMedBERT [11], BlueBERT [28], and a standard BERT base [7]. Each BERT was used to generate embeddings for the train and validation sets, while only the highest performing BERT was used in the test set for final evaluation. Medication text included the name, generic name, class and subclass and past medical history text included code type, name, description along with CCS and description. For both medication and past medical history embeddings, labels were prepended to each column value and at the end of the text for each row, a newline was appended as a divider. The medications and past medical history item length varied a bit for patients, with some having a long

> Medication name: THERAPEUTIC MULTIVITAMIN PO TABS, Generic name: therapeutic multivitamin tablet, Class: MULTIVITAMIN PREPARATIONS, Subclass: Multivitamins
>
> —
>
> Code Type: Dx10, Code: R531, Code Description: Weakness, CCS: 252.0, CCS Description: Malaise and fatigue

Figure 2: Example med and pmh text prior to embedding, added labels are blue

list of items while others had none, which could also be missing data rather than an absence of medications or events. Our model only captures the first 512 tokens for medication and past medical history text respectively to keep runtime and compute cost low.

3.3 Model Definition and Training

Once the data was gathered, cleaned, and enhanced, custom Python scripts to define and train the model were created using the PyTorch library [27]. Prior to model training, embeddings for the train, validation, and test set were generated by the selected BERT models using the HuggingFace transformers library [37] and then saved to files for quick training and evaluation runs. All other features were loaded directly from csv files created during feature selection.

Starting with the embeddings, each of which are represented by a 768 dimension vector for each token, the embedding passes through a 768 by 50 dense layer followed by ReLU activation and batch normalization. After the initial layers for the individual embeddings, all three are combined and concatenated with the raw and engineered features for the first combined hidden layer with an output dim of 64 followed again by ReLU activation and batch normalization. Two more dense-activation-normalization blocks follow with dropout occurring after the second hidden layer until a final dense layer results in two output logits. The final logits are then passed for cross-entropy loss allowing the gradients to flow backwards through the model and update the parameters before the AdamW optimizer [21] takes a new step.

For every epoch, loss and and receiver operating characteristic area under the curve (AUC) were calculated for monitoring. Plateau, cosine annealing [20], and one cycle LR [31] schedulers were configured for identifying an optimal learning rate with cosine annealing providing the fastest convergence and AUC. The model tended

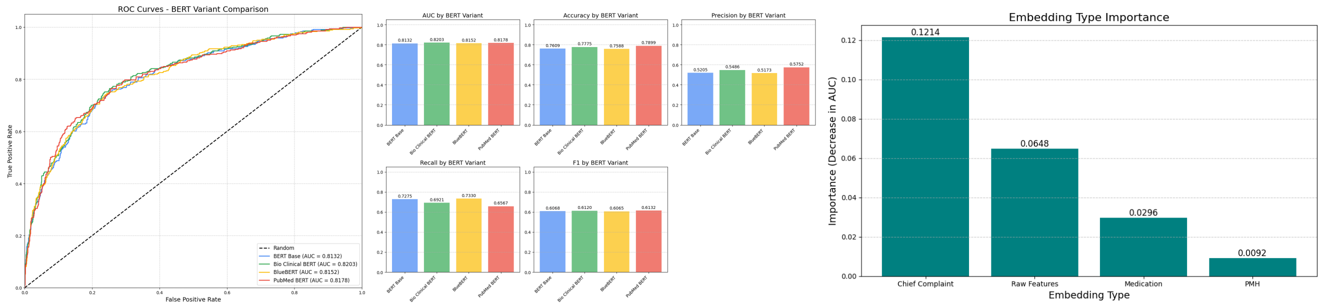

Figure 3: Fast-track prediction test set results for all BERT variants and feature importance comparison for each feature type

to converge after between 10-15 epochs the model used for test evaluation ran for 13 epochs.

3.4 Evaluation

Our model was evaluated against the random test split of MC-MED using the weights from the model with the highest AUC score from training with 4 models total, one for each BERT variant. ROC-AUC is used to evaluate overall performance, since the FT and non-FT classes are highly imbalanced (11,793 under FT criteria versus 105,773 under non-FT criteria). The decision output by the model was compared against our own custom fast-track criteria since MC-MED does not indicate if there is a fast-track available or a patient was admitted there. Based on recent fast-track research [30], we considered an ESI acuity score of 4 or 5 to be fast-tracked and an acuity 3 if the patient LOS was one hour or less.

4 RESULTS

Results with models across BERT variants were close in nearly all metrics as per Table 2. The model using BioClinical BERT achieved the highest across-the-board AUC score at 0.8203 and was followed closely by the PubMed BERT model with 0.8178. In Figure 3 we see that for lower false positive rates (0.05-0.2) the PubMed BERT model has a higher true positive rate than the other variants, but as the false positive rises this effect diminishes. The BERT Base model scored the lowest across all metrics except recall where it surpassed both BioClinical and PubMed BERT. The PubMed BERT model led accuracy, precision, and F1 score metrics but not by any great margin.

To analyze feature importance in the highest AUC model, the model was run again on the training dataset without gradient updates using permutation importance where the model is run with each feature getting randomized values and the performance is compared against the overall performance. Through this analysis we found that triage heart rate, SBP, DBP and age were tabular features that contributed the most to AUC score. We also compared the normal model results to those with permutations where one of the embeddings or all of the raw features together were disabled and found that zeroing the chief complaint embedding had a significant negative impact on the AUC, almost double that of the permutation without the raw features.

5 DISCUSSION

Since an AUC score in the 0.8-0.9 range can be considered an excellent discrimination for a classifier [12], the results show that inferring the fast-track determination for an incoming ED patient is viable, and that additional relevant modalities enhance the performance of the model. The model shared by this paper is one that, with some additional front-end development, could be used by triage staff to aid in their acuity and fast-track decision-making or even potentially by patients before they make a decision to present their condition at the ED. Such a system would enable triage staff to keep the patient flow high which has been shown to lead to lower wait times, length of stay and higher patient satisfaction scores [6, 13, 25, 29]. The high importance of chief complaint shows that meaningful context can be extracted even when data are unstructured text.

One consideration not covered by this paper is the staff allocation and resource availability of an ED. EDs are dynamic, fast moving environments where a few minutes could be critical the life or death of a patient, a concept referred to as the golden hour [4]. If an ED is short a physician or nurse, or a Mass Casualty Incident (MCI) such as a mass shooting or environmental disaster occurs, the possibility of fast-tracking any patients could be dropped in order to preserve precious resources like beds, IVs or staff attention. Such events do not devalue the pursuit of models that can effectively predict acuity or fast-track eligibility, but instead raise additional considerations that make these systems more useful in the real world.

5.1 Bias and Limitations

The tight coupling of our model with MC-MED dataset for training and evaluation introduces a few different biases that are important to be aware of. Firstly, this dataset comes from only one hospital in California, US during a short 3-year period. There could be differences in patients, staff, and procedures depending on the country the hospital operates due to differences in medical laws, hospital organization, or culture and the same goes for different states or regions within the US. The ESI triage system used by the hospital that the MC-MED collected data for to track acuity scores is also subject to bias since other triage systems are used by different hospitals and this could impact the patients considered for fast-track.

The year period 2020-2022 is subject to biases as well since it covers the height of the coronavirus pandemic, an unprecedented international health crisis that significantly impacted health care

Table 2: Performance Comparison of BERT Variants on Test Set

Model	AUC	Accuracy	Precision	Recall	F1
BERT Base	0.8132	0.7609	0.5205	0.7275	0.6068
Bio Clinical BERT	0.8203	0.7775	0.5486	0.6921	0.6120
BlueBERT	0.8152	0.7588	0.5173	0.7330	0.6065
PubMed BERT	0.8178	0.7899	0.5752	0.6567	0.6132

facilities around the world [1]. Throughout this period, EDs were often overloaded as medical staff were dealing with an influx of coronavirus patients, and patients with less severe issues typically opted to skip the ED altogether [24]. Patients eligible for fast track would likely be more hesitant to visit the ED during this period so there could be less FT-specific data for a model to learn from compared to a different period of time. This event also means that the some of the free-text data such as chief complaint could be more biased towards coronavirus like conditions such as respiratory problems.

In terms of demographics, females are represented in a larger distribution than males (53.7% versus 46.1%). For race, white and unknown patients make up the majority of patients and non-hispanic patients have a larger representation for ethnicity.

5.2 Future Work

Our work shows the effectiveness of multi-modal data for fast-track prediction provides abundant opportunity to expand the breadth and depth of this research. Our embeddings were limited to 512 tokens, and some patients medication and past medical history text went well beyond this limit. We imposed this limit since larger embeddings take longer and use more compute. Future work could expand this maximum so all relevant unstructured text relating to a patient is captured.

The fast-track criteria we used could be expanded with finer grained rules such as presence of certain conditions like chest pain or a specific pain level that could be present in a different dataset. More research should be done on the fast-track criteria that triage staff use at different hospitals so that this could be part of a dataset or at the very least, determined from other column values. Staffing and resource allocation information, such as beds available, is critical in whether or not fast-track eligibility is even considered, and future work would greatly benefit from this data.

The data used for input could also be expanded to additional modalities, for example, a patient could provide an audio recording of their symptoms or a picture relevant to their condition, both adding context similar to what the triage staff has available when they make a determination. Future work would need to identify methods of generating audio or image embeddings, and extend the architecture to handle and learn from that input. If patients had some way of putting this information in a secure phone app that could take vitals from a smart device for ED visit recommendations, and the ED could potentially see a lesser load of non-urgent patients leading to the better treatment for higher acuity patients, and feedback that more accurately reflects those that need the ED most.

6 CONCLUSION

As more emergency departments introduce fast-track areas due to their myriad benefits, there is still room to explore the staffing and eligibility of such areas to push benefits even further. The AUC of our model shows that this determination can be made somewhat accurately without post-triage information like acuity and continuous vitals, and features like chief complaint, age, and triage-time vitals are key to an FT-determination. The model presented in this paper is a step towards giving triage staff additional tools that could help make an FT determination and opens the door to an exploration of what other modalities might prove valuable for both the patients considering walking into the ED and the staff in the department that provide care.

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